

Adaptive Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

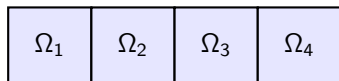
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The Reaction-Diffusion Master Equation [Erban et al. 2007]

Setting. Domain $\Omega \subset \mathbb{R}^d$ partitioned into K voxels of side length h . N species with diffusion coefficients $D_1, \dots, D_N$. State: x_{ik} = molecules of species i in voxel k ; $\mathbf{x}_k = (x_{1k}, \dots, x_{Nk})$ = local voxel state.

diffusion: rate $d_i = D_i/h^2$ per molecule



reactions: propensity $\alpha_r(\mathbf{x}_k)$ in each voxel

$$\frac{dp}{dt} = A_{\text{RDME}} p = \left(\sum_k A_{\text{rxn}}^{(k)} + \sum_{\substack{k, k' \\ \text{adjacent}}} A_{\text{diff}}^{(kk')} \right) p$$

The Computational Bottleneck: FSP [Munsky & Khammash 2006]

The **Finite State Projection** (FSP) cuts the infinite state space $\mathbb{Z}_{\geq 0}^{N \times K}$ down to a finite active set $\mathcal{S}$ and solves the resulting ODE exactly. $n_{\max}$ = how many molecules we allow per species per voxel.

State space size: $|\mathcal{S}| \sim (n_{\max} + 1)^{N \cdot K}$

N	K	$n_{\max}$	$ \mathcal{S} $
1	4	12	$13^4 = 28,561$ tractable
1	8	12	$13^8 \approx 8 \times 10^8$ intractable
1	16	12	$13^{16} \approx 7 \times 10^{17}$ hopeless

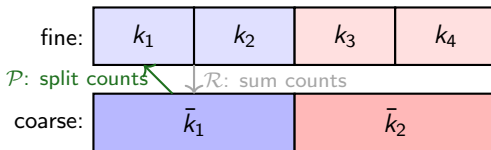
Goal

Exploit the *spatial structure* of the RDME to reduce computation to a sequence of small, coarse-level FSP solves.

Lumpability (Kemeny & Snell): if a set of states transition among themselves *very fast*, you can collapse them into a single macro-state. The internal distribution settles to a known stationary form — that's what we use as the prolongation operator.

Idea: merge pairs of adjacent fine voxels into one coarse voxel. Coarse count for species i , coarse voxel j :

$$\bar{x}_{ij} = x_{i,2j-1} + x_{i,2j}$$



Lumpability parameter (ratio of inter- to intra-pair diffusion rates):

$$\epsilon = \frac{d_{\text{inter}}}{d_{\text{intra}}} = \frac{D/(2h)^2}{D/h^2} = \frac{1}{4} \quad \text{per level (uniform grid)}$$

Small $\epsilon \Rightarrow$ coarsening is accurate.

Restriction and Prolongation

Restriction $\mathcal{R}$: go from fine to coarse by summing over all fine states that share the same coarse counts.

$$(\mathcal{R} p)(\bar{\mathbf{x}}) = \sum_{\mathbf{x}: x_{i,2j-1} + x_{i,2j} = \bar{x}_{ij} \ \forall i,j} p(\mathbf{x})$$

Prolongation $\mathcal{P}$ [Kemeny & Snell 1960; Simon & Ando 1961]: go from coarse back to fine. With $n = \bar{x}_{ij}$ total molecules in coarse voxel j , fast intra-voxel diffusion drives the split to stationarity:

$$\pi(m \mid n) = \text{Bin}\left(n, \frac{1}{2}\right), \quad m = 0, 1, \dots, n$$

(detailed balance; equal-volume sub-voxels). Then:

$$(\mathcal{P} \bar{p})(\mathbf{x}) = \bar{p}(\bar{\mathbf{x}}) \cdot \prod_{i,j} \pi(x_{i,2j-1} \mid \bar{x}_{ij})$$

Consistency: $\mathcal{R} \mathcal{P} = I$

If you go coarse then fine, you get back exactly where you started.

Coarse-Level Generator

Petrov–Galerkin projection: $\bar{A} = \mathcal{R} A_{\text{RDME}} \mathcal{P}$, where

$A_{\text{RDME}} = A_{\text{rxn}} + A_{\text{intra}} + A_{\text{inter}}$:

- A_{intra} : diffusion within each fine pair (fast; rate D/h^2)
- A_{inter} : diffusion between coarse voxels (slow; rate $D/(2h)^2$)

Term	Result
$\mathcal{R} A_{\text{intra}} \mathcal{P}$	$= 0$ (Bin is stationary for A_{intra} ; fast intra-voxel diffusion drops out)
$\mathcal{R} A_{\text{inter}} \mathcal{P}$	Coarse diffusion at rate $D/(2h)^2$, same structural form
$\mathcal{R} A_{\text{rxn}} \mathcal{P}$	Exact for linear (0th- and 1st-order) reactions; error $O(1/n)$ for nonlinear

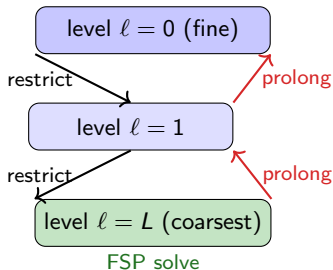
Nice structure

$\bar{A}$ is itself an RDME on a coarser grid — so you can keep coarsening recursively all the way down.

Split $A_{\text{RDME}} = A_{\text{fast}} + A_{\text{slow}}$, where $A_{\text{fast}} = A_{\text{intra}}$ and $A_{\text{slow}} = A_{\text{rxn}} + A_{\text{inter}}$.

V-Cycle: $p(t) \rightarrow p(t + \Delta t)$

- 1 Pre-smooth:** $\tilde{p} \leftarrow e^{\tau A_{\text{fast}}} p$
Run fast diffusion for time τ so molecules even out within each merged pair
- 2 Restrict:** $p^c \leftarrow \mathcal{R} \tilde{p}$
Collapse to the coarse level
- 3 Coarse solve (FSP):** $\bar{p}^c \leftarrow e^{\Delta t \bar{A}} p^c$
Take a full time step on the much smaller coarse problem
- 4 Prolongate correction:**
 $p \leftarrow \tilde{p} + \mathcal{P}(\bar{p}^c - p^c)$
Fold the coarse correction back into the fine solution
- 5 Post-smooth:** $p \leftarrow e^{\tau A_{\text{fast}}} p$



Why this helps:

The coarse level isn't stiff — you can take steps of size $\Delta t \sim 1/\|A_{\text{slow}}\|$ instead of the tiny fine-level limit $\sim 1/\|A_{\text{fast}}\|$.

Speedup $\sim d_{\text{intra}}/k_{\text{rxn}} \gg 1$

Total inter-voxel flux between adjacent voxels k and k' :

$$\Phi_{kk'} = \sum_{i=1}^N d_i \mathbb{E}[x_{ik}] \quad (\text{expected molecules crossing per unit time})$$

Simple rule: merge if $\Phi_{kk'}$ is small. But this breaks in two ways:

- ① **Background inactivity.** Almost no molecules around, so $\Phi_{kk'} \approx 0$ no matter what d_i is. **Fine to merge.**
- ② **Spatial bottleneck.** Molecules are there, but d_i is just slow (membrane, narrow channel). $\Phi_{kk'}$ looks small, but this pair is a gateway between two regions. **Merging here is wrong.**

Fix: refine pair $\{k, k'\}$ if either:

$$\epsilon_{kk'} \geq \epsilon_{\text{lump}} \quad \text{or} \quad \phi_{kk'} = \frac{\Phi_{kk'}}{\sum_i \mathbb{E}[x_{ik}]} \geq \epsilon_{\text{bn}} \cdot \bar{\phi}$$

where $\bar{\phi}$ = mean per-molecule flux across the domain; $\epsilon_{\text{lump}}, \epsilon_{\text{bn}}$ are user thresholds.

Benchmark 1: Birth-Death-Diffusion (V-Cycle, $K = 4$)

Model: $\emptyset \xrightarrow{k_b} A \xrightarrow{k_d} \emptyset$, $k_b = 2$, $k_d = 1$, $D = 1$. **Smooth IC:** product-Poisson with $\mu = [4, 2, 1, 1]$ per voxel.

$K = 4$ fine, $K = 2$ coarse. Full V-cycle with Binomial- π prolongation. Ground truth: π^* (product-Poisson stationary distribution).

Metric	Fine FSP	MG V-cycle
$TV(p, \pi^*)$ at $t = 0.5, 1, 2, \dots$	$\searrow 0$	0.000 at all t
State space $ \mathcal{S} $	$\sim 10,400$	~ 190
Wall time	5.3 s	0.22 s

Exact coarse generator for linear networks

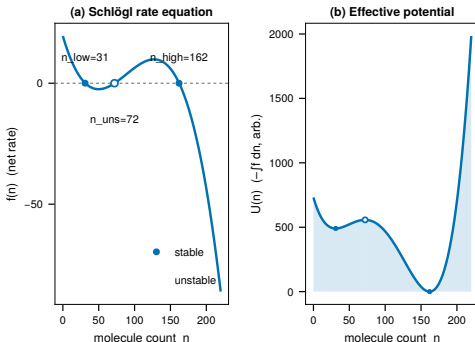
For birth-death-diffusion the Petrov–Galerkin term $\mathcal{R}A_{\text{rxn}}\mathcal{P}$ is *exact*. The V-cycle therefore matches π^* **perfectly** at every time step.

54 $\times$ fewer states. 24 $\times$ faster.

Benchmark 2: Spatial Schlögl Model

Model: $2A + B \rightleftharpoons 3A$, $\emptyset \rightleftharpoons A$ (species B held fixed).

Bistable: the rate equation $f(n)$ has three fixed points; the two stable ones sit at $n_{\text{low}} = 31$ and $n_{\text{high}} = 162$.



Left: rate equation $f(n)$; zeros are fixed points. **Right:** potential $U(n) = -\int_0^n f \, dm$; wells at $n = 31, 162$; barrier at $n = 72$.

Three Prolongation Approaches

Coarse variable $n_c = n_1 + n_2$ loses directionality. Three ways to reconstruct $p(n_1 | n_c)$:

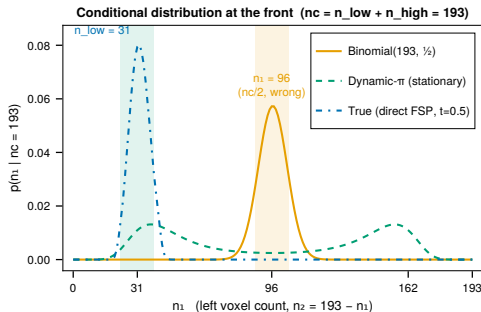
Binomial- π : $\pi(n_1 | n_c) = \text{Bin}(n_c, \frac{1}{2})$. Free to compute; ignores all reaction structure. **Symmetric — peaks at $n_1 = n_c/2$.**

Dynamic- π : True stationary conditional of the $K = 2$ RDME, $\pi^*(n_1 | n_c)$, precomputed once. Bistable: *bimodal* at $(n_{\text{low}}, n_{\text{high}})$ and $(n_{\text{high}}, n_{\text{low}})$ — still 50/50 weight. **Symmetric — can't distinguish which voxel holds which state.**

Fix-3 (fine-conditional injection): For states already represented in the fine distribution, rescale multiplicatively (inject, don't split). For newly created coarse states, fall back to $\pi^*(\cdot | n_c)$. **Preserves the directionality of the front.**

Prolongation Failure: The Symmetry Trap

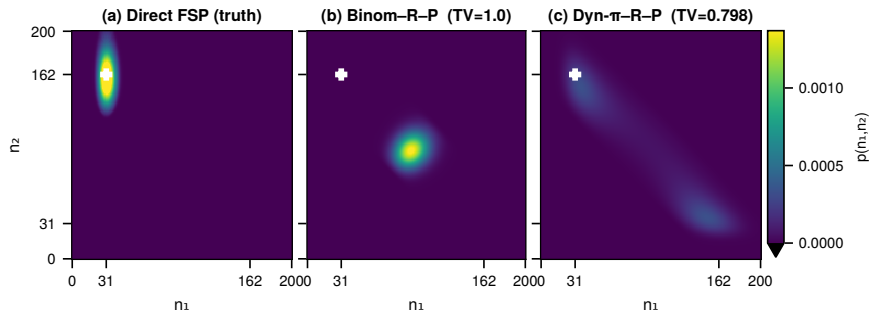
Setup: $K = 2$, IC $(n_1, n_2) = (31, 162)$. Coarse variable $n_c = 193$.



- **Binomial- π :** $\text{Bin}(193, \frac{1}{2})$ peaks at $n_1 = 96$ (midpoint). TV= 1.000.
- **Dynamic- π :** Stationary conditional $\pi^*(n_1 | 193)$ bimodal at 31 and 162, but 50/50 weight (stationary is symmetric under $n_1 \leftrightarrow n_2$). TV= 0.798.

Symmetry trap: $n_c = 193$ is consistent with both $(31, 162)$ and $(162, 31)$; the coarse variable loses the directionality of the front.

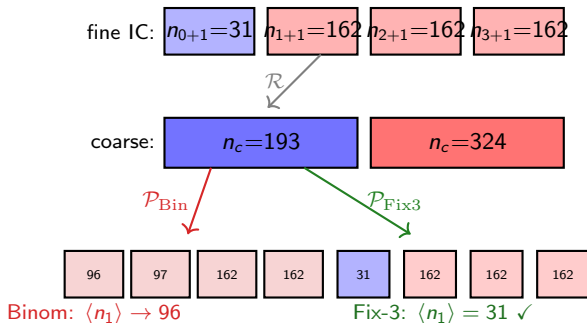
2D Joint Distribution at $t = 0.5$



- (a) Direct FSP (ground truth): mass at $(n_1, n_2) = (31, 162)$.
(b) Binomial- π : mass at $(96, 97)$ on the anti-diagonal. TV= 1.
(c) Dynamic- π : bimodal at $(31, 162)$ and $(162, 31)$. TV= 0.798 (symmetry trap).

Fix-3: Fine-Conditional Prolongation

Key insight: the fine distribution before the coarse step already encodes which voxel is in which state. Use it directly instead of re-splitting via Binomial.



Implementation: for each fine state (n_1, n_2) already in $\mathcal{S}$, multiply its probability by the coarse correction ratio $\bar{p}^c(n_c)/p^c(n_c)$. No Binomial enumeration — state space stays *bounded*.

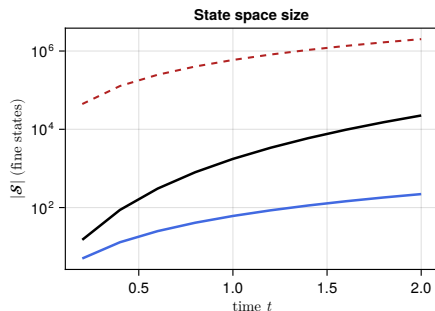
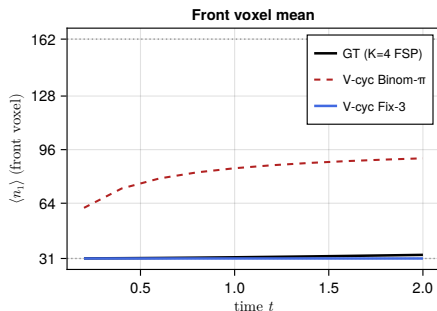
Asymmetric Front: $K = 2$ (Step 1)

IC: $(n_1, n_2) = (31, 162)$. Ground truth: direct $K = 2$ FSP ($\sim 40,000$ states).

Method	$\langle n_1 \rangle$	$\langle n_2 \rangle$	TV $t=0.5$	TV $t=2$	Time
Direct FSP	31.0	162.0	—	—	ref
V-cyc Binom- π	≈ 95	≈ 98	0.999	0.999	$1\times$
V-cyc Fix-3	31.0	162.0	0.98	0.98	$5\times$ faster

- **Binom- π** : immediately loses the front ($n_c = 193 \rightarrow$ midpoint 96).
- **Fix-3**: correctly tracks $\langle n_1 \rangle = 31$ at all times; 5 to 51 fine states (vs 40k for FSP).
- Residual TV reflects the coarse-level approximation, not directionality loss.

Asymmetric Front: $K = 4$ — Results



IC: (31, 162, 162, 162). **Left:** $\langle n_1 \rangle$ (front voxel mean) vs time. Fix-3 tracks the front exactly; Binom- π immediately loses directionality. **Right:** fine state space size on log scale. Fix-3: $|\mathcal{S}| \lesssim 221$; Binom- π : $|\mathcal{S}| \rightarrow 2 \times 10^6$.

Asymmetric Front: $K = 4$ (Step 2)

IC: (31, 162, 162, 162). Front in coarse pair $\{1, 2\}$: $n_c^{(1)} = 193$.

Ground truth: $K = 4$ adaptive FSP (evolving state space).

Method	$\langle n_1 \rangle$	$\langle n_2 \rangle$	$\langle n_3 \rangle$	$\langle n_4 \rangle$	Time	$ \mathcal{S} $
GT ($K = 4$ FSP)	31.0	162.0	162.0	162.0	1.2 s	22,551
V-cyc Binom- π	≈ 90	≈ 103	162	162	65.2 s	2,000,000+
V-cyc Fix-3	31.0	162.0	162.0	162.0	0.18 s	221

Fix-3 dominates on all axes

360 $\times$ faster than Binom- π . **6 $\times$ faster** than ground truth. **100 $\times$ smaller** state space than GT.

Binom- π : state space explodes exponentially (recursive Binomial splits);
 $K = 4 \rightarrow 2$ requires enumerating all pairwise combinations.

Asymmetric Front: $K = 8$ (Step 3)

IC: $(31, \underbrace{162, \dots, 162}_7)$. Front in coarse pair $\{1, 2\}$ after $K = 8 \rightarrow K = 4$ restriction.

Method	$\langle n_1 \rangle$	$\langle n_2 \rangle$	$\langle n_3 \rangle$	$\langle \dots \rangle$	Time
$K = 8$ direct FSP	—	—	—	—	intractable ($>75k$ states at $t=$
V-cyc Binom- π	—	—	—	—	state space explodes in prok
V-cyc Fix-3	31.0	162.0	162.0	162.0	1.7 s

Scalability: Fix-3 fine state count grows modestly ($K = 2$: 51, $K = 4$: 221, $K = 8$: 8361) while Binom- π explodes factorially.

Why Binom- π explodes at $K = 8$

Each coarse voxel n_c spawns $n_c + 1$ fine splits; with $K/2 = 4$ coarse voxels, the recursion generates $O(n_c^4)$ fine states per coarse state. Fix-3 never enumerates Binomial combinations at all.

Method

- Spatial coarsening via lumpability [Kemeny & Snell 1960];
 $\epsilon = d_{\text{inter}}/d_{\text{intra}} = 1/4$ per level
- V-cycle [Briggs et al. 2000]: smooth $\rightarrow$ restrict $\rightarrow$ coarse FSP $\rightarrow$ prolong $\rightarrow$ smooth
- Three prolongation operators: Binom- π , Dyn- π , Fix-3
- **Fix-3**: multiplicative injection preserves directionality of spatial fronts
- Two-criterion adaptive refinement: lumpability + per-molecule bottleneck

Results

- Birth-death: $\text{TV}(p_{\text{MG}}, \pi^*) = 0$ at all times (exact coarse generator); **24 $\times$ faster**, 54 $\times$ fewer states
- Schlögl $K = 4$: Fix-3 **360 $\times$ faster** than Binom- π , **6 $\times$ faster** than direct FSP; 221 vs 2M states
- Schlögl $K = 8$: Fix-3 solves in **1.7 s**, 8361 states; direct FSP and Binom- π both intractable
- Binom- π / Dyn- π : symmetry trap, $\text{TV} \approx 1$ for asymmetric fronts

Open

- Multi-level Fix-3
($K \rightarrow K/2 \rightarrow K/4 \rightarrow \dots$)
- Adaptive spatial refinement

- Erban, Chapman, Maini (2007). A practical guide to stochastic simulations of reaction-diffusion processes. *arXiv:0704.1908*.
- Munsky & Khammash (2006). The finite state projection algorithm for the solution of the chemical master equation. *J. Chem. Phys.*, 124:044104.
- Kemeny & Snell (1960). *Finite Markov Chains*. Van Nostrand.
- Simon & Ando (1961). Aggregation of variables in dynamic systems. *Econometrica*, 29:111–138.
- Briggs, Henson, McCormick (2000). *A Multigrid Tutorial*, 2nd ed. SIAM.
- Dendukuri, Chandrasekaran, Petzold (2025). Flux-preserving adaptive FSP for multiscale stochastic reaction networks. *SIAM Multiscale Model. Simul.* (under review).